Dalibor Buljovčić - Portfolio

ENTREPRENEURIAL

PLANAR DAB TRANSFORMER

48V > 800V system ; 3.6kW SiC based System
100kHz Operation with Triple Phase shift DAB

Characteristics:

- Bobbinless transformer with external leakage inductance
- Gapped N95 ER/PLT core magnetic structure
- Custom 6mm/3oz PCB with concatenated hole contacts

Customer Service:

- Technical project/stakeholder requirement management
- Converter architecture consulting and system engineering
- Electric and mechanical transformer elements design
- Functional converter simulation in **Plecs** and **Matlab**
- Magnetic and Thermal FEM simulation in **Ansys**
- Supplier management and prototyping

POWER ELECTRONICS AND MAGNETICS

QUADRATIC-BOOST CONVERTER - FUEL CELL APPLICATION

***100kW GaN based* DC/DC converter platform development for Hydrogen Fuel Cell Systems**

Characteristics:

- Non-isolated Quadratic-Boost with high voltage gain
- High-Frequency Coupled Inductor Magnetics array

Responsibilities:

- Power core product segment ownership
- Market research, project potential and development cost estimation
- Converter architecture and mechanical concept
- Functionality scope and proof of concept
- Supply chain and technology scoping

DUAL STAGE DC/AC CONVERTER - REEFER SUPPLY

2x15kW DAB converter integrated with 22kW 3-phase Grid forming Inverter

Characteristics:

- 15kW two chamber winding transformer with flux focus elements
- Flat winding FeSi powder resin molded PFC choke array

Responsibilities:

- Power core product development leadership
- EMC device concept and power conversion architecture
- Technical sourcing and supplier management
- FEM simulation

EMC FUNCTION OWNERSHIP - E-MOBILITY DIVISION

OBC EMC filtering solutions

- 22kW segmented wire-wound racetrack design
- DC and AC side conducted emission filtering concept

Traction Inverter EMC filtering solutions

- 300kW multi-turn laser welded winding CMC with integrated filtering capacitors
- Multi stage filtering configuration for commercial and passenger vehicle segment

SYSTEM

Booster Pump IoT controller

- Frequency regulated 1kW PSM motor controller with PFC stage
- Digital electronics design and IoT connectivity

Drain Pump IoT controller

- IoT plafrom controller for pump operation monitoring and water presence detection
- Consumer style package with plug and HMI - mechatronics

DIGITAL ELECTRONICS

SMARTPHONE ACCESS AUTOMOTIVE ECU

- Technical responsibility for hardware end-end development
- ECU Hub module for vehicle antenna coomunication acquisition
- Design cordination and management of technical team
- Automotive industrialisation and development processes

DATA LOGGER INTERFACE BOARD FOR AUTONOMOUS PARKING SYSTEM

- High-Speed HDI PCB design of DDR2 memory
- Ethernet 100Mb/s + Broad-Reach
- USB and CAN communication

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